

V. Imp #

Hasse Diagram

Hasse Diagram is a ^{representation - used} ~~visual~~ drawing in mathematics to show the ordering of elements in a Partial Order Set (POSET) in the clean and simple way.

Steps To Design Hasse Diagram:

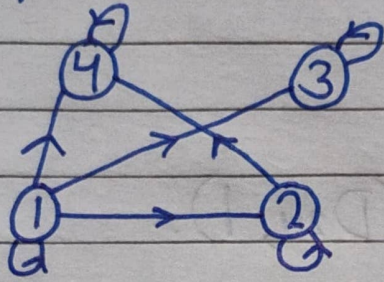
1. Start with a POSET-set of item and a valid ordering rule (like number divisibility or subset).
2. Remove self links & transitive links - Ignore the collection where items relate to itself and if $A \rightarrow B$ & $B \rightarrow C$ then remove the direct link between $A \rightarrow C$.
3. Draw & Arrange - Place smaller elements at the bottom and larger elements at the top, draw straight lines between elements where one immediately covers the other (elements/items are shown as a dot).

Here, arrowheads are not shown because direction is implicitly understood as moving upwards.

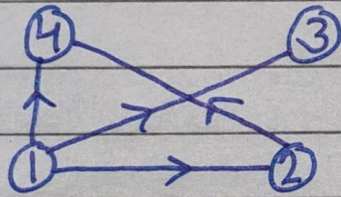
Q.1) Draw Hasse Diagram for $A = (\{1, 2, 3, 4\}, \leq)$ with relation $R = \{(1, 1), (1, 2), (1, 3), (1, 4), (2, 2), (2, 4), (3, 3), (4, 4)\}$.

Solⁿ:

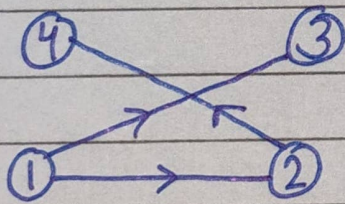
Create graph,



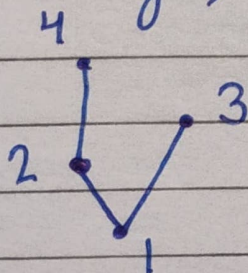
Remove self links,



Remove transitive links,



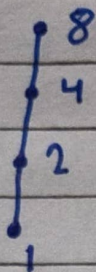
Draw & Arrange,



Q.2) Draw Hasse Diagram for $(D_8, |)$.

Solⁿ:

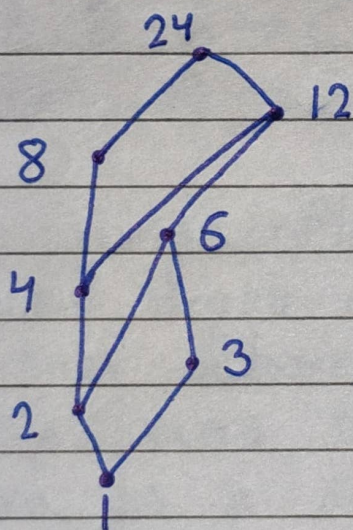
Set of divisors of 8 is $\{1, 2, 4, 8\}$ i.e. D_8 .



Q.3) Draw diagram for $(D_{24}, |)$.

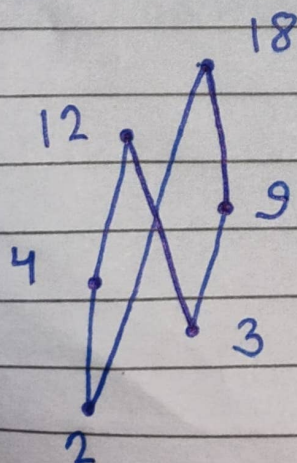
Solⁿ:

$D_{24} = \{1, 2, 3, 4, 6, 8, 12, 24\}$

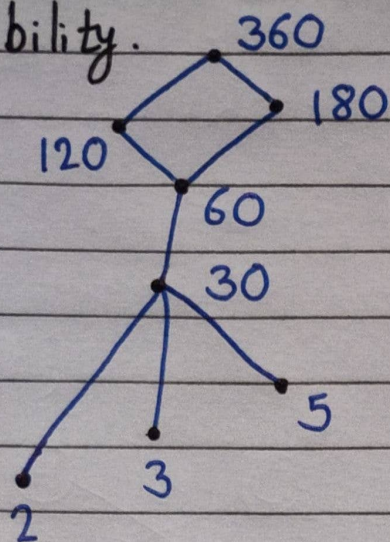


Q.4) Draw diagram for $A = \{2, 3, 4, 9, 12, 18\}$ under $|$.

Solⁿ:



Q.5) Draw diagram for $A = \{2, 3, 5, 30, 60, 120, 180, 360\}$ under divisibility.
Solⁿ:



Q.6) Draw hasse diagram for power set $(P(A), \subseteq)$ where $A = \{a, b, c\}$.
Solⁿ:

$$P(A) = \{\emptyset, \{a\}, \{b\}, \{c\}, \{a, b\}, \{a, c\}, \{b, c\}, \{a, b, c\}\}$$

